

Homework 3

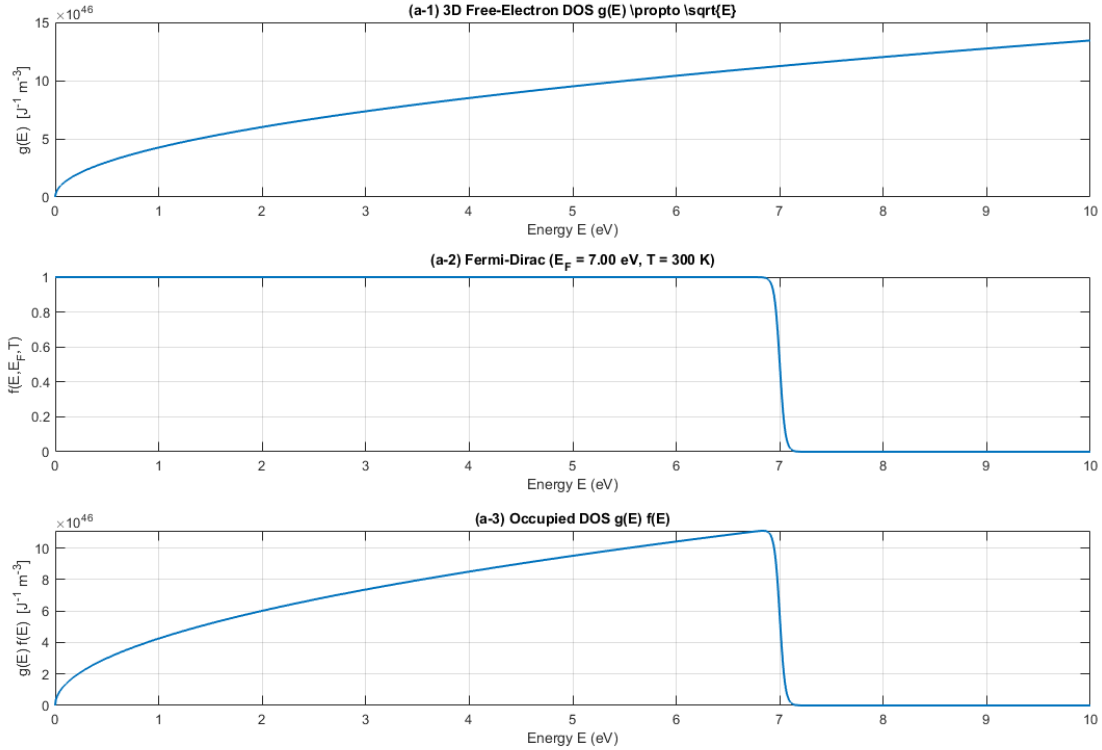
PROBLEM: FREE-ELECTRON METAL ($m^* = m_e$). DOS, FERMI ENERGY, AND VELOCITIES

Throughout we use the 3D free-electron density of states per unit volume and per unit energy

$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} \sqrt{E}, \quad [g] = \text{J}^{-1} \text{m}^{-3},$$

the Fermi-Dirac probability $f(E, \mu, T) = [1 + \exp((E - \mu)/k_B T)]^{-1}$, and the occupied DOS $g(E)f(E, \mu, T)$. Energies shown on plots are in eV, integrals are done in SI (J).

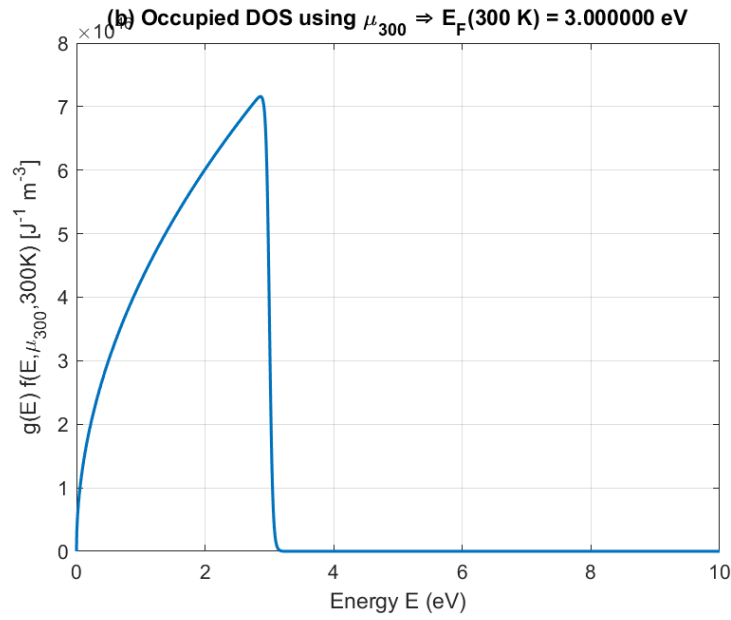
Question 1. (a) Plots of $g(E)$, $f(E, E_F, T)$, and $g(E)f$ for $E_F = 7 \text{ eV}$, $T = 300 \text{ K}$.



$g(E)$ increases as $\sqrt{E}$ because the number of k -states on a shell scales as $k^2 \propto E$. $f(E)$ controls occupancy; their product shows how many states are actually filled at each energy.

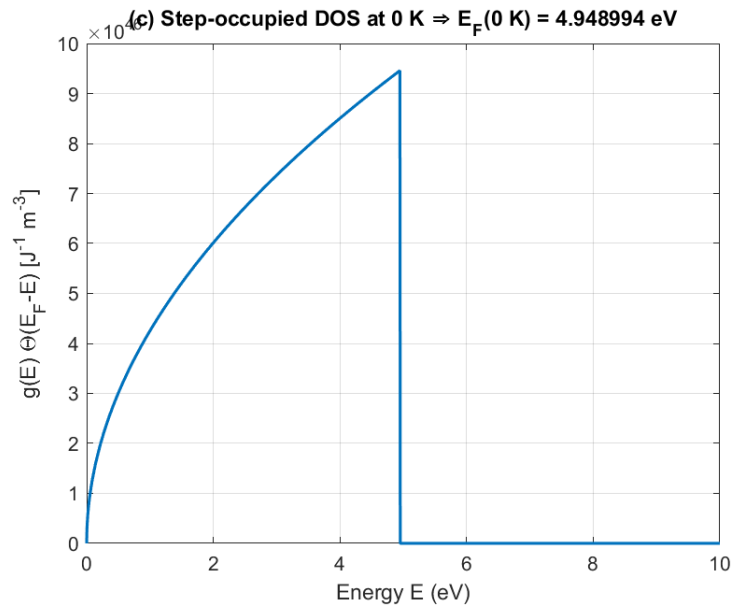
Question 2. (b) Copper with $n \simeq 5 \times 10^{28} \text{ m}^{-3}$. Find E_F at 300 K from $n = \int g(E)f(E, \mu, T) dE$ over 0–10 eV. Numerically solving $F(\mu) = \int_0^{10 \text{ eV}} g(E)f(E, \mu, 300 \text{ K}) dE - n = 0$ yields

$$E_F(300 \text{ K}) \approx 4.948883 \text{ eV}.$$



Question 3. (c) Repeat at $T = 0 \text{ K}$. At $T = 0$, f is a step: $n = \int_0^{E_F(0)} g(E) dE \Rightarrow E_F(0) = \frac{\hbar^2}{2m_e} (3\pi^2 n)^{2/3}$. For $n = 5 \times 10^{28} \text{ m}^{-3}$,

$$E_F(0) \approx 4.948994 \text{ eV}.$$



Question 4. (d) Compare E_F at 300 K and 0 K.

$$\Delta E_F \equiv E_F(300 \text{ K}) - E_F(0) \approx \boxed{-0.111 \text{ millieV}} \ll k_B T|_{300 \text{ K}} \approx 25.9 \text{ millieV}.$$

Thus the $T = 0$ expression is a very good approximation for room-temperature metals.

Question 5. (e) Electron velocities and average energies at 300 K and 0 K. With $v(E) = \sqrt{2E/m_e}$,

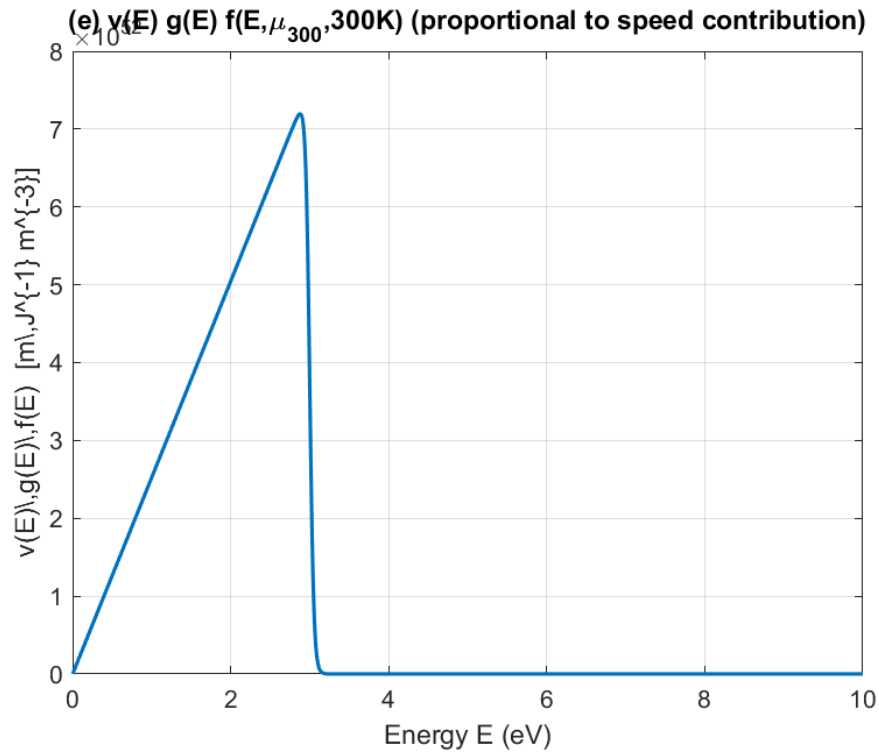
$$\langle v \rangle = \frac{1}{n} \int_0^\infty v(E) g(E) f(E, \mu, T) dE, \quad \langle E \rangle = \frac{1}{n} \int_0^\infty E g(E) f(E, \mu, T) dE.$$

At $T = 0$: $v_F = \sqrt{2E_F(0)/m_e}$, $\langle v \rangle = \frac{3}{4}v_F$, $\langle E \rangle = \frac{3}{5}E_F(0)$. Numerically (from the script):

$$v_F \approx 1.319 \times 10^6 \text{ m/s}, \quad \langle v \rangle_{0 \text{ K}} \approx 9.896 \times 10^5 \text{ m/s}, \quad \langle v \rangle_{300 \text{ K}} \approx 9.896 \times 10^5 \text{ m/s},$$

$$\langle E \rangle_{0 \text{ K}} \approx 2.9694 \text{ eV}, \quad \langle E \rangle_{300 \text{ K}} \approx 2.9697 \text{ eV}.$$

Only a fraction $\sim k_B T/E_F \sim 10^{-3}$ of electrons near E_F are thermally perturbed, so averages change negligibly.



Question 6. (f) Why do electrons (almost) maintain their average energy as $T \rightarrow 0$? Is their randomness maintained? By the Pauli exclusion principle, all single-particle states are filled up to E_F even at $T = 0$. Thermal randomness vanishes as $T \rightarrow 0$, but quantum randomness remains: electrons occupy a broad distribution of momenta up to k_F with zero net drift. Hence typical speeds/energies stay finite (set by E_F), while thermal fluctuations disappear.

6.1. APPENDIX

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1
2 %% Anne, Gautham HW 3: EE381
3 clear; clc; close all;
4
5 % Physical constants
6 eV = 1.602176634e-19; % [J]
7 kB = 1.380649e-23; % [J/K]
8 hbar = 1.054571817e-34; % [J*s]
9 me = 9.1093837015e-31; % [kg]
10
11 % Helper functions
12 gJ = @(E) (1/(2*pi^2)) * (2*me/hbar^2)^(3/2) .* sqrt(max(E,0)); % DOS [J^-1 m
    ^-3]
13 fFD = @(E,mu,T) 1./(1 + exp((E - mu)./(kB*T))); % Fermi-Dirac
14 vE = @(E) sqrt(2*max(E,0)/me); % speed [m/s]
15
16 % Energy grid for plotting -(010 eV)
17 Emin_eV = 0; Emax_eV = 10; NE = 2000;
18 E_eV = linspace(Emin_eV, Emax_eV, NE);
19 E_J = E_eV * eV;
20
21 % (a) DOS, Fermi function, occupied DOS for EF=7 eV, T=300 K
22 EF_assumed_eV = 7.0; T_plot = 300;
23 muJ_assumed = EF_assumed_eV * eV;
24
25 g_plot = gJ(E_J);
26 f_plot = fFD(E_J, muJ_assumed, T_plot);
27 gf_plot = g_plot .* f_plot;
28
29 figure('Name','(a) DOS, Fermi-Dirac, and Occupied DOS','NumberTitle','off');
30 tiledlayout(3,1);
31 nexttile; plot(E_eV, g_plot,'LineWidth',1.5); grid on;
32 xlabel('E (eV)'); ylabel('g(E) [J^{-1} m^{-3}]'); title('(a-1) DOS g(E) \proptopto
    \surd E');
33 nexttile; plot(E_eV, f_plot,'LineWidth',1.5); grid on;
34 xlabel('E (eV)'); ylabel('f(E,E_F,T)'); title(sprintf('(a-2) F-D (E_F=%.1f eV,
    T=%d K)',EF_assumed_eV,T_plot));
35 nexttile; plot(E_eV, gf_plot,'LineWidth',1.5); grid on;
36 xlabel('E (eV)'); ylabel('g(E)f(E) [J^{-1} m^{-3}]'); title('(a-3) Occupied DOS
    g(E)f(E)');
37
38 %% (b) Copper density and EF at 300 K (robust)
39 n_target = 5e28; % [m^-3]
40 T300 = 300; % [K]
41
42 % Analytic EF at 0 K
43 EF0_J = (hbar^2/(2*me)) * (3*pi^2*n_target)^(2/3);
44 EF0_eV = EF0_J / eV;
45
46 % Use a single finite integration cutoff everywhere (b)-(e)
47 % 80 kT beyond mu is plenty; enforce >= 50 eV for safety.
48 kT_J = kB*T300;
49 Emax_J_int_min = 50*eV;

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50 Emax_J_int = max(EF0_J + 80*kT_J, Emax_J_int_min);
51
52 % Number-density integral and residual
53 integrand_n = @(E,mu,T) gJ(E).*fFD(E,mu,T);
54 F_mu = @(mu) integral(@(E) integrand_n(E,mu,T300), 0, Emax_J_int, ...
55                     'RelTol',1e-8,'AbsTol',1e-14) - n_target;
56 % Initial guess..
57 kT_eV = kB*T300/eV;
58 dmu_somm_eV = -(pi^2/12)*(kT_eV^2)/(EF0_eV);
59 mu_guess = (EF0_eV + dmu_somm_eV)*eV;
60
61 try
62     mu300_J = fzero(F_mu, mu_guess);
63 catch
64     scan_eV = linspace(EF0_eV-0.02, EF0_eV+0.02, 81); % ±20 meV window
65     Fvals = arrayfun(@(x) F_mu(x*eV), scan_eV);
66     s = sign(Fvals);
67     idx = find(s(1:end-1).*s(2:end) < 0, 1);
68     if isempty(idx), error('Could not bracket root; widen scan.');
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end

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72
73 EF_300_eV = mu300_J / eV;
74
75 % Plot occupied DOS for (b) on -010 eV window
76 gf_300 = gJ(E_J).*fFD(E_J, mu300_J, T300);
77 figure('Name','(b) Occupied DOS at 300 K','NumberTitle','off');
78 plot(E_eV, gf_300,'LineWidth',1.5); grid on;
79 xlabel('E (eV)'); ylabel('g(E)f(E) [J^{-1} m^{-3}]');
80 title(sprintf('(b) Occupied DOS, E_F(300K)=%.6f eV', EF_300_eV));
81
82 % (c) Step-occupied DOS at 0K
83 gf_0K = gJ(E_J).*(E_J<=EF0_J);
84 figure('Name','(c) Occupied DOS at 0 K','NumberTitle','off');
85 plot(E_eV, gf_0K,'LineWidth',1.5); grid on;
86 xlabel('E (eV)'); ylabel('g(E)\Theta(E_F-E)');
87 title(sprintf('(c) Step-occupied DOS, E_F(0K)=%.6f eV', EF0_eV));
88
89 % (d) Comparison
90 dEF_meV = 1e3*(EF_300_eV - EF0_eV);
91 fprintf('(d) E_F(0 K) = %.6f eV\n', EF0_eV);
92 fprintf('    E_F(300 K) = %.6f eV\n', EF_300_eV);
93 fprintf('    Shift ΔE_F = %.3f meV (Sommerfeld ~ %.3f meV)\n\n', ...
94         dEF_meV, dmu_somm_eV*1e3);
95
96 %% (e) Velocities and average energies (use the SAME cutoff)
97 % 0 K (analytic)
98 vF = sqrt(2*EF0_J/me);
99 v_mean_0K = 0.75*vF;
100 E_mean_0K = 0.6*EF0_J;
101
102 % 300 K (numeric with finite cutoff)
103 num_v = integral(@(E) vE(E).*gJ(E).*fFD(E, mu300_J, T300), 0, Emax_J_int, ...
104                 'RelTol',1e-7,'AbsTol',1e-13);

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105 den_n = integral(@(E) gJ(E).*fFD(E, mu300_J, T300),      0, Emax_J_int, ...
106                'RelTol',1e-8,'AbsTol',1e-14);
107 v_mean_300K = num_v / den_n;
108
109 num_E = integral(@(E) E.*gJ(E).*fFD(E, mu300_J, T300),      0, Emax_J_int,
110                ...
111                'RelTol',1e-7,'AbsTol',1e-13);
112 E_mean_300K = num_E / den_n;
113 fprintf('(e) v_F (0 K)    = %.3e m/s\n', vF);
114 fprintf('    <v> (0 K)    = %.3e m/s    (=3/4 v_F)\n', v_mean_0K);
115 fprintf('    <v> (300 K) = %.3e m/s\n', v_mean_300K);
116 fprintf('    <E> (0 K)    = %.4f eV    (=3/5 E_F)\n', E_mean_0K/eV);
117 fprintf('    <E> (300 K) = %.4f eV    ( unchanged)\n\n', E_mean_300K/eV);

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LISTING 1. MATLAB script for HW3